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Pedestrians as obstacles in dense crowds: Local encounters and collective motion

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ABSTRACT

Understanding how different pedestrian movements relate to crowd dynamics is an important topic in pedestrian and evacuation research. However, direct comparisons of pedestrians who remain stationary, move against the main flow or cross it remain limited, particularly regarding their local interactions and relationships with collective motion. Here, we use trajectory data from the Festival of Lights in Lyon to classify pedestrians by walking speed and direction and examine the associated local interactions and crowd-level motion. Counterflow encounters show shorter times to closest approach among approaching pairs and greater local velocity variance than encounters involving stationary or crossing pedestrians. After accounting for density, a higher proportion of these non-mainstream pedestrians is associated with lower corridor-axis alignment and greater crowd-level velocity variance, predominantly in the longitudinal component, whereas associations with mean walking speed are less consistent across recordings. An idealized continuum model further illustrates how prescribed differences in walking direction and speed contribute to these crowd-level responses. These findings provide a clearer understanding of how heterogeneous pedestrian movements reshape local interactions and the organization of dense flow in public spaces.

1. Introduction

Pedestrian crowds are rarely uniform: even when most pedestrians move in the same direction, some may stop, move against the main flow, or cross it Gu et al. (2025). Such heterogeneous motion is common in railway stations, event venues, urban corridors, and other densely occupied public spaces (Yi et al., 2024b), where individual movement can interact with the surrounding crowd and contribute to collective patterns (Yi et al., 2024a; Yi and Wu, 2026). Understanding these collective dynamics is an important topic in crowd science and is also relevant to the safety and operational management of public infrastructures (Helbing et al., 2005). Over the past decades, extensive research has established macroscopic regularities of pedestrian traffic, including the relationships among density, speed, and flow (Zhang et al., 2011; Chraibi et al., 2025). However, real pedestrian streams are often heterogeneous (Wu et al., 2022a,b, 2023), and differences in walking behavior and motion state can be accompanied by lane fragmentation, local congestion, and increased velocity fluctuations (Murakami et al., 2021). Therefore, understanding how different pedestrian motion states are related to both local interactions and collective dynamics remains an important problem.

A considerable amount of research has examined how obstacles and geometric constraints affect pedestrian movement. Experimental studies have investigated pedestrian flows around fixed and temporary obstacles (Wang et al., 2020; Wu et al., 2024), pedestrian

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route choices around obstacles (Li et al., 2019; Wu et al., 2025), and the effects of obstacles near exits (Shiwakoti et al., 2019; Hu et al., 2023). Other studies have considered more complex walking environments, including bottlenecks (Li et al., 2023a), elevated platforms (Song et al., 2025), and merging flows (Fu et al., 2023). In parallel, computational models have been developed to describe pedestrian movement in the presence of obstacles and geometric restrictions. Macroscopic approaches have been used to examine obstacle placement and evacuation performance (Cristiani and Peri, 2017), while microscopic and data-driven approaches have been applied to bidirectional corridors (Li et al., 2023b), T-junctions (Wang et al., 2025), and bottlenecks (Jiang et al., 2023). These studies demonstrate that physical obstacles and environmental geometry can strongly affect pedestrian movement. In real crowds, however, local disturbances can also arise from pedestrians themselves rather than from fixed structures.

Several forms of such pedestrian motion have been studied separately. Stationary pedestrians can restrict the space available to surrounding pedestrians, particularly near exits and other crowded locations (Shiwakoti et al., 2019; Hu et al., 2023). Pedestrians moving against the dominant stream interact directly with oncoming pedestrians, and mutual anticipation can contribute to the formation and stability of bidirectional flow patterns (Murakami et al., 2021; Yi et al., 2023). Crossing pedestrians are common at intersections and in intersecting pedestrian streams, where their trajectories differ substantially from the dominant direction of movement (Rakotoarivelo et al., 2023; Jin et al., 2013; Jain et al., 2014). Large-scale trajectory studies have also provided detailed measurements of pedestrian dynamics and demonstrated how local interactions are related to collective crowd behavior (Corbetta et al., 2014; Appert-Rolland et al., 2014). Nevertheless, stationary, counterflow, and crossing pedestrians are usually considered in different experimental or modeling settings. A direct comparison of these motion states within the same empirical dataset remains limited. In particular, it is still unclear whether they are associated with different local motion patterns and how their prevalence is related to directional alignment, velocity variation, and traffic flow.

To address these questions, this study analyzes pedestrian trajectories recorded during the Festival of Lights in Lyon (Dufour et al., 2025). Pedestrian motion is first classified according to walking speed and direction into mainstream, stationary, counterflow, crossing, and unclassified states. We then compare the motion of mainstream pedestrians near stationary, counterflow, and crossing pedestrians using measures of pairwise separation, walking speed, heading direction, time to closest approach, and local velocity variance. At the macroscopic scale, we examine how the combined proportion of the three non-mainstream states is associated with crowd-level speed, signed flow, alignment with the corridor axis, and velocity variance while accounting for density. Finally, an idealized continuum model is used to examine how differences in pedestrian speed, direction, and proportion contribute to these crowd-level responses. The empirical results show that encounters with counterflow pedestrians have shorter times to closest approach and greater local velocity variance than encounters with stationary or crossing pedestrians. At the crowd level, a higher proportion of stationary, counterflow, and crossing pedestrians is associated with lower alignment with the corridor axis and greater velocity variance, whereas the association with mean walking speed is less consistent across recordings. These findings help clarify how different pedestrian motion states are related to local interactions and the organization of dense flow.

The rest of this paper is organized as follows. Section 2 describes the methods used in the empirical and continuum analyzes. Section 3 presents the empirical and numerical results. Finally, Section 4 discusses the main findings and limitations, summarizes the principal conclusions, and outlines directions for future work.

2. Methods

To identify different pedestrian motion states and examine how they are related to local pedestrian interaction and crowd-level flow, we analyze trajectory data recorded during the Festival of Lights in Lyon (Dufour et al., 2025). The dataset contains 9 camera views, including TopView_1A–1C and TopView_2A–2F, with 5269 recorded tracks. The analysis proceeds in four steps: we first classify pedestrian motion according to walking speed and direction, and then quantify the local motion of mainstream pedestrians near stationary, counterflow, and crossing pedestrians. Next, we examine how the prevalence of these three states is associated with crowd-level speed, flow, directional alignment, and velocity variance. Finally, an idealized continuum model is used to examine how differences in pedestrian speed, direction, and proportion affect the corresponding crowd-level measures under controlled conditions.

2.1. Pedestrian-state classification

We classify pedestrian motion from walking speed and direction. Here, two centered time windows are used for different purposes: a short window of $W_s = 30$ frames (1 s) to estimate walking speed and a long window of $W_l = 90$ frames (3 s) to estimate walking direction. For pedestrian i at frame t, let $S_{it}(W_s)$ denote the observations available in the short window. We calculate the mean walking speed as

$$\bar{v}_{it}^{\text{short}} = \frac{1}{|S_{it}|} \sum_{\tau \in S_{it}} \|\mathbf{v}_{i\tau}\|, \tag{1}$$

where $\mathbf{v}_{i\tau}$ is the instantaneous velocity of pedestrian i at frame τ. The walking direction is estimated from the longer window. Let $L_{it}(W_l)$ denote the available observations in the long window. We calculate this metric as

$$\mathbf{e}_{it}^{\text{long}} = \frac{\bar{\mathbf{v}}_{it}^{\text{long}}}{\|\bar{\mathbf{v}}_{it}^{\text{long}}\|}, \tag{2}$$

where $\bar{\mathbf{v}}_{it}^{\text{long}} = \frac{1}{|L_{it}|} \sum_{\tau \in L_{it}} \mathbf{v}_{i\tau}$ represents the mean walking velocity during the long window.

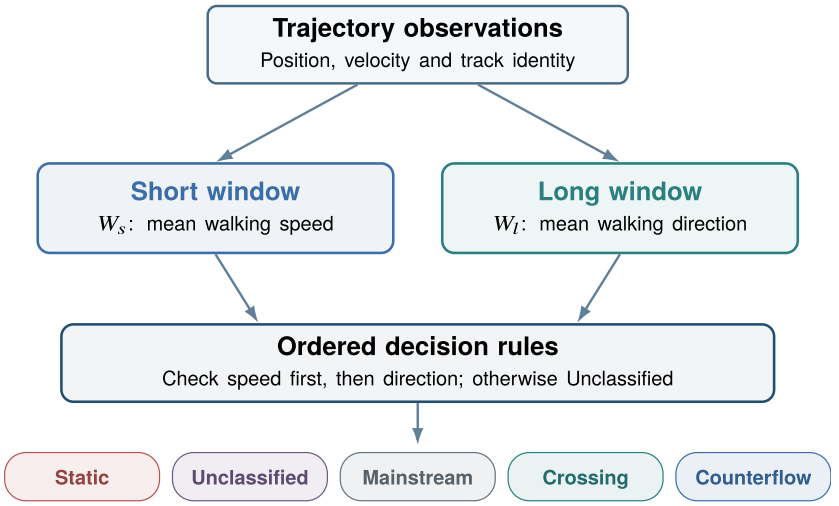


Fig. 1. Workflow for classifying pedestrian motion.

The prescribed reference traffic direction is leftward in the data coordinates, $\mathbf{n} = (-1, 0)$. Given that opposite directions along the same corridor should define the same corridor axis, we estimate a reference axis for each camera view from the second moment of the long-window directions, as given by

$$\mathbf{M}_s = \sum_{i,t} \frac{1}{n_i N_s} \mathbf{e}_{it}^{\text{long}} (\mathbf{e}_{it}^{\text{long}})^T, \tag{3}$$

where n_i is the number of observations from track i that contribute to the scene-axis calculation, and N_s is the number of tracks in camera view s with at least one contributing observation. An observation is included only when the pedestrian is non-static ($\bar{v}_{it}^{\text{short}} \geq 0.10\,\text{m s}^{-1}$), the 90-frame window contains at least 30 observations, and the norm of the long-window mean velocity is finite and greater than $10^{-12}\,\text{m s}^{-1}$.

The eigenvector associated with the largest eigenvalue of $\mathbf{M}_s$ is oriented towards $\mathbf{n}$ and denoted by $\mathbf{e}_s^{\text{ref}}$. The alignment of each observation with this reference direction is then expressed as

$$D_{it} = \mathbf{e}_{it}^{\text{long}} \cdot \mathbf{e}_s^{\text{ref}}. \tag{4}$$

Based on this, an observation can be classified as Static when $\bar{v}_{it}^{\text{short}} < 0.10\,\text{m s}^{-1}$, while a non-static observation without a valid long-window direction is classified as Unclassified. The remaining observations are classified as Mainstream when $D_{it} \geq 0.7$, Crossing when $-0.7 \leq D_{it} < 0.7$, and Counterflow when $D_{it} < -0.7$, in which the directional threshold of 0.7 corresponds to an angular half-width of almost 45°. The specific workflow for classifying pedestrian motion is summarized in Fig. 1.

2.2. Local interaction measures

After identifying pedestrian states, we examine how mainstream pedestrians move in the vicinity of Static, Counterflow, and Crossing pedestrians. At each frame, each Mainstream pedestrian i is paired with the nearest pedestrian j belonging to one of these three states, provided that their center-to-center distance is less than 4 m. Let $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$ denote the relative position from the Mainstream pedestrian to the selected pedestrian. The longitudinal and lateral directions are defined by $\mathbf{n} = (-1, 0)$ and $\mathbf{p} = (0, 1)$, respectively. We first compute the signed longitudinal separation and the absolute lateral separation as

$$\ell_{ij} = \mathbf{r}_{ij} \cdot \mathbf{n}, \qquad d_{ij}^{\perp} = \left| \mathbf{r}_{ij} \cdot \mathbf{p} \right|. \tag{5}$$

Here, $\ell_{ij} > 0$ indicates that the selected pedestrian is ahead of the Mainstream pedestrian along the reference direction, and the quantity $d_{ij}^{\perp}$ measures their lateral center-to-center separation.

We measure the deviation of individual motion direction from the longitudinal direction as

$$\alpha_{it} = \cos^{-1}\left[\frac{\bar{\mathbf{v}}_{it}^{\text{short}} \cdot \mathbf{n}}{\|\bar{\mathbf{v}}_{it}^{\text{short}}\|} \right], \tag{6}$$

where $\bar{\mathbf{v}}_{it}^{\text{short}} = \frac{1}{|S_{it}|} \sum_{\tau \in S_{it}} \mathbf{v}_{i\tau}$ denotes the mean walking velocity during the short window. The angle α_{it} measures the heading deviation from the longitudinal direction.

To examine close interactions, we further retain frames in which the distance between the selected pair is less than 1.5 m. Three additional measures are calculated here: 1) The lateral motion of the Mainstream pedestrian is quantified by the squared lateral

component of its mean velocity, as given by

$$A_{it}^{\perp} = \left(\bar{\mathbf{v}}_{it}^{\text{short}} \cdot \mathbf{p}\right)^2. \tag{7}$$

This quantity measures the magnitude of lateral motion without distinguishing its direction. 2) We calculate the time to closest approach (TCA) under the assumption that both pedestrians maintain their current mean velocities (Ge and Wu, 2026). Let $\mathbf{u}_{ij} = \bar{\mathbf{v}}_{jt}^{\text{short}} - \bar{\mathbf{v}}_{it}^{\text{short}}$ be their relative velocity. For approaching pairs satisfying $\|\mathbf{u}_{ij}\| > 10^{-12}\,\text{m s}^{-1}$ and $\mathbf{r}_{ij} \cdot \mathbf{u}_{ij} < 0$, the time to closest approach and the corresponding minimum separation are

$$\tau_{ij}^{\text{TCA}} = -\frac{\mathbf{r}_{ij} \cdot \mathbf{u}_{ij}}{\|\mathbf{u}_{ij}\|^2}, \qquad d_{ij}^{\text{TCA}} = \left\|\mathbf{r}_{ij} + \tau_{ij}^{\text{TCA}}\mathbf{u}_{ij}\right\|. \tag{8}$$

3) Local differences in pedestrian velocities are measured within a circular neighborhood of radius $R_{\text{loc}} = 2.5$ m centered on the Mainstream pedestrian. The neighborhood $\eta_i(t)$ contains all pedestrians with valid positions and mean velocities, irrespective of their motion state. With $\bar{\mathbf{v}}_{\eta_i} = \frac{1}{|\eta_i(t)|}\sum_{k\in\eta_i(t)} \bar{\mathbf{v}}_{kt}^{\text{short}}$, the local velocity variance is defined as

$$\sigma_{v,i}^2 = \frac{1}{|\eta_i(t)|}\sum_{k\in\eta_i(t)} \left\|\bar{\mathbf{v}}_{kt}^{\text{short}} - \bar{\mathbf{v}}_{\eta_i}\right\|^2. \tag{9}$$

This reflects differences in both walking speed and direction among nearby pedestrians.

2.3. Crowd-level measures

After characterizing local interactions, we quantify crowd-level motion at two spatial scales. Pedestrian density, walking speed, and signed flow are calculated within a fixed $1\,\text{m} \times 1\,\text{m}$ measurement region, whereas the combined proportion of Static, Counterflow, and Crossing pedestrians, corridor-axis alignment, and velocity variance are calculated over the entire camera view.

For fixed-region measures, the measurement region R is defined by $x \in [-1.5m, -0.5m]$ and $y \in [3.5m, 4.5m]$ in the data coordinates. Let V_i denote the Voronoi cell of pedestrian i and $|V_i \cap R|$ the area of that cell falling inside R. The Voronoi density and walking speed (Zhang et al., 2011) are defined as

$$\rho_R = \frac{1}{|R|}\sum_i \frac{|V_i \cap R|}{|V_i|}, \qquad v_R = \frac{1}{|R|}\sum_i \bar{v}_{it}^{\text{short}}|V_i \cap R|. \tag{10}$$

Only frames in which finite Voronoi cells fully cover the measurement region are retained. With $u_{n,i} = \bar{\mathbf{v}}_{it}^{\text{short}} \cdot \mathbf{n}$, the signed flow along the reference direction is calculated as

$$J_n = \frac{1}{|R|}\sum_i u_{n,i}\frac{|V_i \cap R|}{|V_i|}. \tag{11}$$

Here, $J_n > 0$ indicates a net pedestrian flow along the prescribed leftward direction, whereas $J_n < 0$ indicates it in the opposite direction.

We next describe the composition and velocity organization of the entire camera view. The combined proportion of Static, Crossing, and Counterflow pedestrians is defined as

$$R_{\text{obs}} = \frac{N_{\text{Static}} + N_{\text{Crossing}} + N_{\text{Counterflow}}}{N_{\text{total}}}, \tag{12}$$

where N_{total} is the total number of pedestrians. Thus, R_{obs} represents the fraction of all tracked pedestrians in a frame that belong to one of the three non-mainstream states.

The alignment with the corridor axis is quantified by

$$S_{\parallel} = \frac{1}{N_{\text{ns}}}\sum_{i=1}^{N_{\text{ns}}} \cos(2\theta_i), \tag{13}$$

where θ_i is the angle between the short-window mean velocity and the prescribed horizontal axis, and N_{ns} is the number of non-static pedestrians. The factor $2\theta_i$ treats motion in either direction along the corridor axis as aligned: $S_{\parallel} = 1$ for purely longitudinal motion in either direction and $S_{\parallel} = -1$ for purely transverse motion.

The velocity differences among pedestrians in the entire camera view are quantified by

$$\Phi_v = \frac{1}{N_{\text{total}}}\sum_{i=1}^{N_{\text{total}}} \left\|\bar{\mathbf{v}}_{it}^{\text{short}} - \bar{\mathbf{v}}_t^{\text{short}}\right\|^2, \tag{14}$$

where $\bar{\mathbf{v}}_t^{\text{short}} = \frac{1}{N_{\text{total}}}\sum_i \bar{\mathbf{v}}_{it}^{\text{short}}$ is the mean velocity vector of all pedestrians in the frame. Equivalently, it is the sum of the population variances of the longitudinal and lateral velocity components. The opposing streams can therefore produce a large Φ_v even when each stream is internally ordered.

We then examine the relationship between the combined non-mainstream proportion and each crowd-level outcome while accounting for pedestrian density. A separate model is fitted within each camera view:

$$Y_{sf} = \beta_{0s} + g_s(\rho_{R,sf}) + \beta_{\mathrm{obs},s}\frac{R_{\mathrm{obs},sf} - \overline{R}_{\mathrm{obs},s}}{0.10} + \epsilon_{sf}, \tag{15}$$

Here, s indexes the camera view, f indexes the frame, Y_{sf} is the crowd-level outcome, β_{0s} is the view-specific intercept, and ϵ_{sf} is the residual error. The function $g_s(\rho)$ is a restricted cubic spline of the fixed-region density ρ_R, with knots at the 5th, 35th, 65th, and 95th density percentiles of that view. Besides, R_{obs} is a whole-view predictor, $\overline{R}_{\mathrm{obs},s}$ is the mean obstacle ratio in view s, and the coefficient $\beta_{\mathrm{obs},s}$ represents the associated change in Y for a 0.10 increase in R_{obs} within view s, after accounting for density.

2.4. Continuum model

To complement the empirical analysis, we further use a continuum model to examine how differences in pedestrian proportion, direction, and mobility affect crowd-level motion under controlled conditions. The model considers Mainstream pedestrians together with one of the three non-mainstream states, namely Static, Counterflow, or Crossing pedestrians.

Let $\rho(x, y, t)$ denote the total pedestrian density and $\phi(x, y) \in [0, 1]$ the prescribed local proportion of the non-mainstream state. The evolution of pedestrian density follows the conservation equation (Bain and Bartolo, 2019; Hughes, 2002)

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \overline{\mathbf{u}}\right) = 0, \tag{16}$$

where $\overline{\mathbf{u}}$ is the mean velocity of the mixed pedestrian flow

$$\overline{\mathbf{u}} = (1 - \phi)\mathbf{u}_{\mathrm{m}} + \phi\mathbf{u}_c. \tag{17}$$

The velocity of Mainstream pedestrians is defined as

$$\mathbf{u}_{\mathrm{m}} = K(\phi)s(\rho)\mathbf{n}. \tag{18}$$

The reduction in pedestrian mobility associated with an increasing proportion of non-mainstream pedestrians is defined as $K(\phi) = 1 - (1 - K_{\min})\phi$, where $K_{\min}$ controls the strength of this reduction. The walking speed is assumed to decrease linearly with density according to $s(\rho) = v_{\mathrm{f}}\left(1 - \rho/\rho_{\mathrm{jam}}\right)$, where $v_{\mathrm{f}} = 1.05\,\mathrm{m\,s^{-1}}$ is the free walking speed and $\rho_{\mathrm{jam}} = 3.66\,\mathrm{m^{-2}}$ is the jam density. For a non-mainstream state c, the corresponding velocity is defined as

$$\mathbf{u}_c = \eta_c K(\phi)s(\rho)\mathbf{e}_c. \tag{19}$$

The three non-mainstream states differ in their prescribed directions and speeds. Static pedestrians are assigned zero velocity, $\mathbf{u}_c = \mathbf{0}$. Counterflow pedestrians move in the opposite direction, $\mathbf{e}_c = (1, 0)$, whereas Crossing pedestrians move in the transverse direction, $\mathbf{e}_c = (0, 1)$. The relative speed factors are estimated from the empirical data as $\eta_{\mathrm{counterflow}} = 1.016$ and $\eta_{\mathrm{crossing}} = 0.549$.

The conservation equation is solved using a finite-volume method on a primary 60×30 grid. The numerical flux is calculated by the first-order local Lax–Friedrichs/Rusanov numerical flux (LeVeque, 2002), as given by the equation

$$\widehat{F}(\rho^-, \rho^+) = \frac{F(\rho^-) + F(\rho^+)}{2} - \frac{a}{2}(\rho^+ - \rho^-), \tag{20}$$

where ρ^- and ρ^+ denote the density states on the two sides of a cell interface. For each interface, F denotes the local density flux in the corresponding coordinate direction, evaluated using the prescribed ϕ field, and a is an upper bound on the corresponding characteristic speed. The time step is then chosen according to the characteristic-speed bounds in both coordinate directions:

$$\Delta t\left(\frac{a_x}{\Delta x} + \frac{a_y}{\Delta y}\right) \le C_{\mathrm{CFL}}, \tag{21}$$

where $C_{\mathrm{CFL}} = 0.45$ and $\Delta t \le 0.05\,\mathrm{s}$. Here, a_x and a_y are the corresponding characteristic-speed bounds in the longitudinal and transverse directions.

3. Results

3.1. Identifying pedestrian obstacles in dense flows

This section reports the identification results of obstacles (i.e., individuals whose motion patterns locally disrupt the mainstream flow, rather than barriers composed of specific objects) in dense flows, based on the empirical dataset provided by Dufour et al. (2025). Based on the classification framework introduced in the previous section, each pedestrian's specific trajectory segment can be categorized into one of five pedestrian types: Mainstream, Static, Unclassified, Crossing, and Counterflow. Unlike conventional crowd analysis that focuses primarily on global characteristics, this identification emphasizes behavioral heterogeneity as the key mechanism through which pedestrian obstacles emerge and affect collective movement.

Fig. 2 visualizes the spatial distribution of the identified pedestrian types in nine typical footage, in which Fig. 2(a)–(c) represent TopView_1 (TopView_1A-1C) and Fig. 2(d)–(i) correspond to TopView_2 (TopView_2A–2F). Across these scenarios, Mainstream (gray) pedestrians form the dominant flow, whereas obstacles appear as localized clusters or streaks of non-mainstream behaviors.

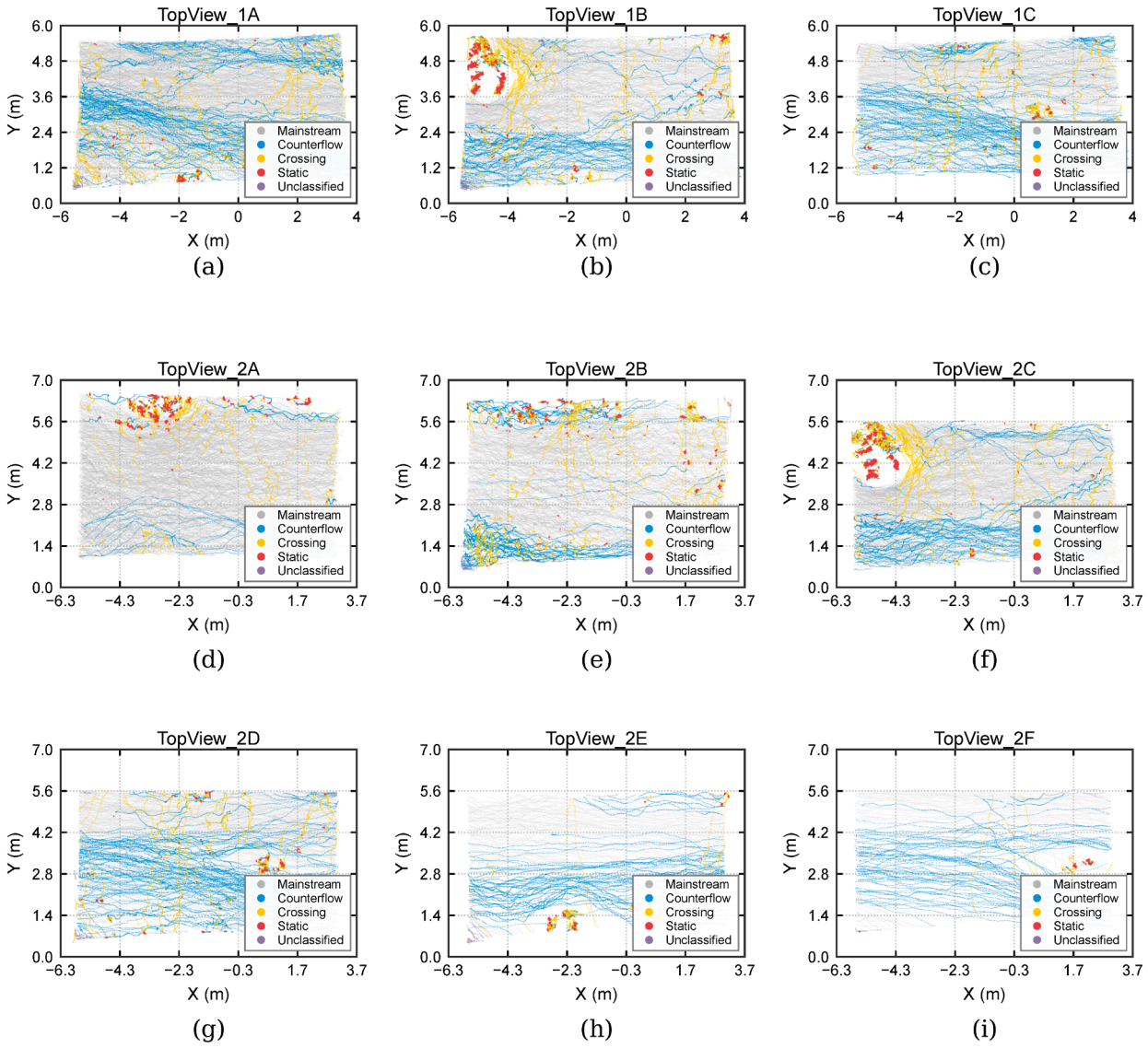


Fig. 2. Spatial distribution of pedestrian motion states. (a–i) TopView_1A–1C and TopView_2A–2F. Each point represents a classified pedestrian–frame record in the data coordinates. TopView_1B and TopView_2C are overlapping camera views of the same physical scene.

Specifically, pedestrian obstacles exhibit a mixed pattern in which Static (red) clusters appear near the boundary while Counterflow (blue) and Crossing (yellow) pedestrians are scattered throughout the corridor, indicating that different obstacle mechanisms can coexist within the same flow. In TopView_1B and TopView_2C, Static pedestrians concentrate in a compact region near the left side, producing an evident “frozen” obstacle area around which the Mainstream stream bends and compresses. In contrast, TopView_2E and TopView_2F show a clear structure where Counterflow pedestrians form a persistent counter-flow band, producing a long-range disturbance rather than a point-like blockage. Overall, these identification results provide a diverse and behavior-rich dataset, which enables systematic quantification of how different obstacle types and ratios reshape pedestrian flow patterns in subsequent analyzes.

To move beyond a qualitative analysis, as shown in Fig. 3, we summarize the statistical signatures of different pedestrian motion states in TopView_1 and TopView_2. The proportion distributions in Fig. 3(a) and (d) show that Mainstream pedestrians occupy the largest share in most cases, typically concentrated at high proportions, whereas Crossing, Counterflow, Static, and Unclassified pedestrians remain minority components with distributions strongly skewed toward low values. Despite their small proportions, these minority types are non-negligible because they represent concentrated local disturbances rather than uniformly distributed noise. The speed distributions in Fig. 3(b) and (e) reveal clear separation between behavioral groups: Mainstream and Counterflow pedestrians generally maintain moderate-to-high walking speeds, while Crossing pedestrians tend to move more slowly and Static pedestrians concentrate near zero speed, consistent with their interpretation as moving and stationary obstacles. Finally, the direction-angle distributions in Fig. 3(c) and (f) provide the strongest validation of the identification: Mainstream pedestrians cluster near the

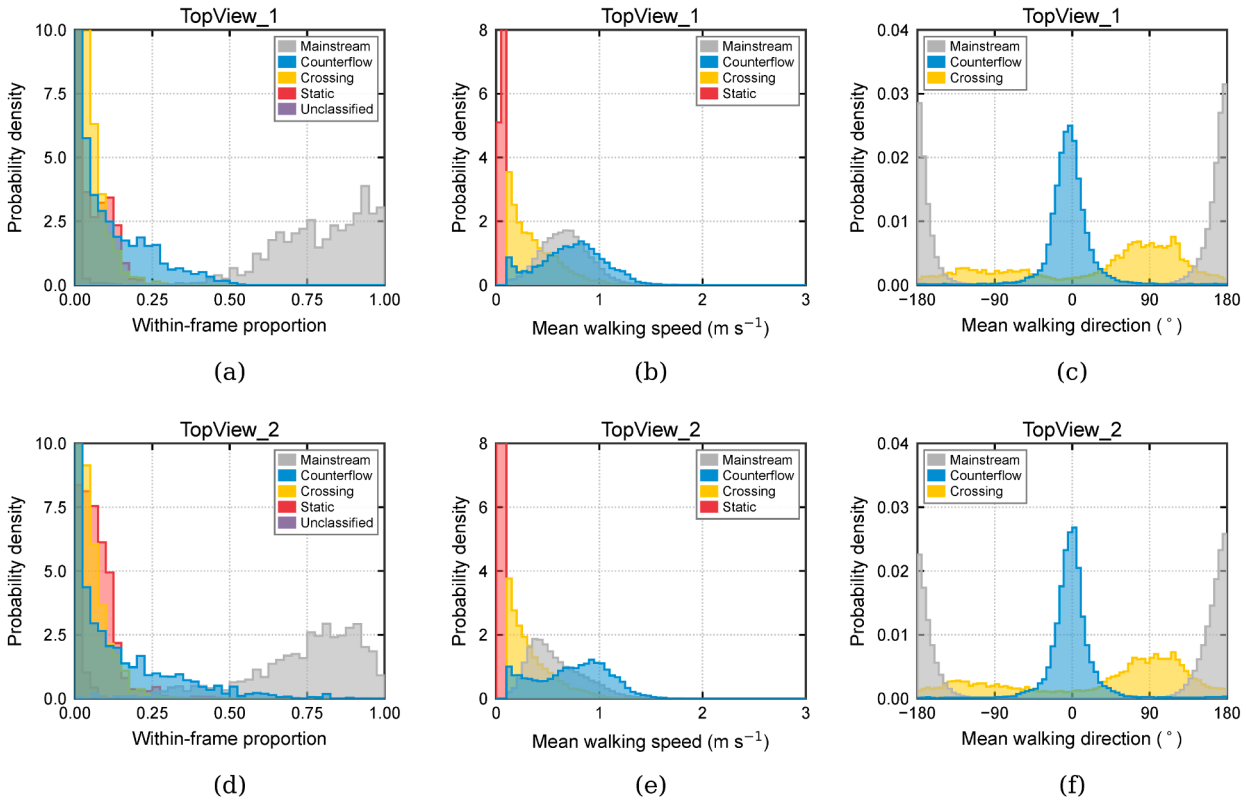


Fig. 3. Distributions of pedestrian motion states. (a, d) Distributions of within-frame proportions of the five states for TopView_1 and TopView_2. (b, e) Distributions of the mean walking speed. (c, f) Distributions of the mean walking direction. Histograms are normalized within each state and camera group.

dominant flow direction (near ±180°), Counterflow pedestrians concentrate around the opposite direction (about 0°), and Crossing pedestrians form an intermediate peak around vertical angles (around ±90°). More importantly, TopView_1 and TopView_2 exhibit highly consistent statistical patterns, suggesting that the identified obstacle categories are stable across cameras and scenes, and thus suitable for comparative impact analysis in later sections.

3.2. Local motion near different pedestrian obstacles

To quantify how different pedestrian obstacles are related to local motion, we examine the response of Mainstream pedestrians near Static, Counterflow, and Crossing pedestrians. Fig. 4(a)–(c) first compares the relative positions, walking speeds, and heading deviations of the selected pairs as a function of longitudinal separation. The lateral distance between the paired pedestrians is generally larger near zero longitudinal separation in Fig. 4(a), but its possible range becomes narrower at large longitudinal separations. For differences among the three states within the same longitudinal bins, Static pedestrians show the largest lateral separation in most supported bins, followed by Counterflow and Crossing pedestrians. Mean walking speed is also lower near zero longitudinal separation than in the outer part of the profile for all three states in Fig. 4(b). Compared with the near-zero region, the corresponding speed difference is 0.049 m s^{-1} for Static, 0.108 m s^{-1} for Counterflow, and 0.095 m s^{-1} for Crossing. Different longitudinal-separation bins may contain different interaction events, so this profile should not be interpreted as the same pedestrian slowing down over the course of one encounter. Heading deviation does not show the same ordering among the three states in Fig. 4(c): in the two bins adjacent to zero longitudinal separation, the mean deviations are 14.41° for Static, 12.58° for Counterflow, and 15.61° for Crossing.

To compare close interactions, we then consider pairs separated by less than 1.5 m. The squared lateral speed differs little among the three states in Fig. 4(d): mean values are 0.0349 m^2 s^{-2} for Counterflow, 0.0410 m^2 s^{-2} for Crossing, and 0.0360 m^2 s^{-2} for Static, with the 95% CIs for all paired differences across the nine camera views including zero. In contrast, time to closest approach differs markedly in Fig. 4(e). Among approaching pairs, mean TCA is shortest for Counterflow (0.414 s; 95% CI: 0.313 to 0.516 s), followed by Crossing (1.687 s; 95% CI: 1.313 to 2.061 s) and Static (2.037 s; 95% CI: 1.724 to 2.350 s). However, the predicted minimum center separation 0.885 m for Counterflow is the largest, compared with 0.742 m for Crossing and 0.733 m for Static, showing that a shorter TCA does not imply a smaller minimum separation. Local velocity variance also differs among the states in Fig. 4(f), reaching 0.402 m^2 s^{-2} for Counterflow, compared with 0.239 m^2 s^{-2} for Crossing and 0.232 m^2 s^{-2} for Static. Counterflow has the highest variance, consistent with the coexistence of pedestrians moving in opposite longitudinal directions. Omitting TopView_1B changes the displayed

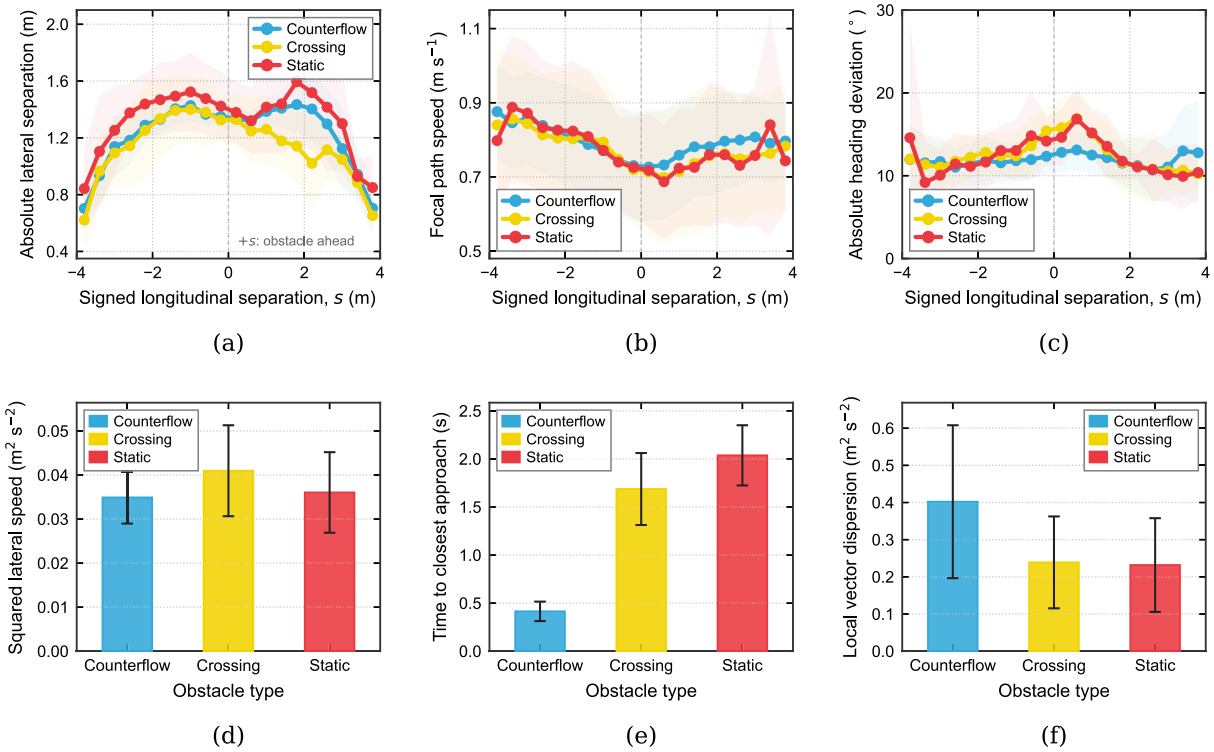


Fig. 4. Local motion near Static, Counterflow, and Crossing pedestrians. (a–c) Absolute lateral separation, focal path speed, and absolute heading deviation as functions of signed longitudinal separation. (d–f) Measures for pairs separated by less than 1.5 m: squared lateral speed, time to closest approach (TCA), and local velocity variance. Curves in panels (a–c) and bar heights in panels (d–f) are first averaged within interaction events and then equally weighted across contributing camera views. Shading and error bars show 95% Student-t intervals across views.

bin means in Fig. 4(a)–(c) by at most 0.078 m, 0.0173 m s^{-1}, and 1.61°, respectively. Removing the documented seven-person Static group and reconstructing all pairings changes squared lateral speed by at most 0.0010 m^2 s^{-2}, TCA by 2.43%, and local velocity variance by 0.64%. Overall, Counterflow is distinguished mainly by shorter approach times and larger local velocity differences rather than stronger lateral motion.

3.3. Crowd-level relationships with non-mainstream pedestrians

In this section, we examine how the proportion of non-mainstream (Static, Counterflow, and Crossing) pedestrians is related to crowd-level motion. Given that non-mainstream states can coexist within the same frame, their prevalence is summarized by R_{obs}. Fig. 5 presents the relationships between R_{obs} with mean walking speed, signed flow, corridor-axis alignment, and velocity variance.

As shown in Fig. 5(a), a 0.10 increase in R_{obs} is associated with a mean walking-speed change of −0.0211 m s^{-1} (95% CI: −0.0483 to 0.0061 m s^{-1}). As the confidence interval includes zero, the average association with walking speed remains uncertain. In Fig. 5(d), signed flow shows a clearer negative association, changing by −0.0658 m^{-1} s^{-1} (95% CI: −0.1177 to −0.0139 m^{-1} s^{-1}) per 0.10 increase in R_{obs}. Across the supported density range, signed flow increases from low to medium density, whereas its behavior at higher densities remains uncertain; we therefore do not infer a descending high-density branch. The time-series examples in Fig. 5(b) and (e) illustrate that periods of higher R_{obs} are generally accompanied by lower corridor-axis alignment within individual camera views, although quantitative inference is based on the density-adjusted models. Across the nine view-specific models, $S_{\parallel}$ decreases by 0.0393 (95% CI: −0.0621 to −0.0165) per 0.10 increase in R_{obs}. The binned summaries in Fig. 5(c) further show that velocity variance is generally higher at larger R_{obs} across much of the supported density range, while Fig. 5(f) shows this relationship more directly. After adjustment for density, velocity variance grows by an average of 0.0806 m^2 s^{-2} (95% CI: 0.0580 to 0.1032 m^2 s^{-2}) per 0.10 increase in R_{obs}. Most of this increase occurs in the longitudinal velocity component, whereas the lateral contribution is much smaller. The associations with alignment and velocity variance also remain robust to alternative treatments of the overlapping TopView_1B and TopView_2C views. Overall, higher R_{obs} is most consistently associated with lower corridor-axis alignment and greater velocity variance, whereas its association with mean walking speed remains uncertain and the negative association with signed flow is clearest.

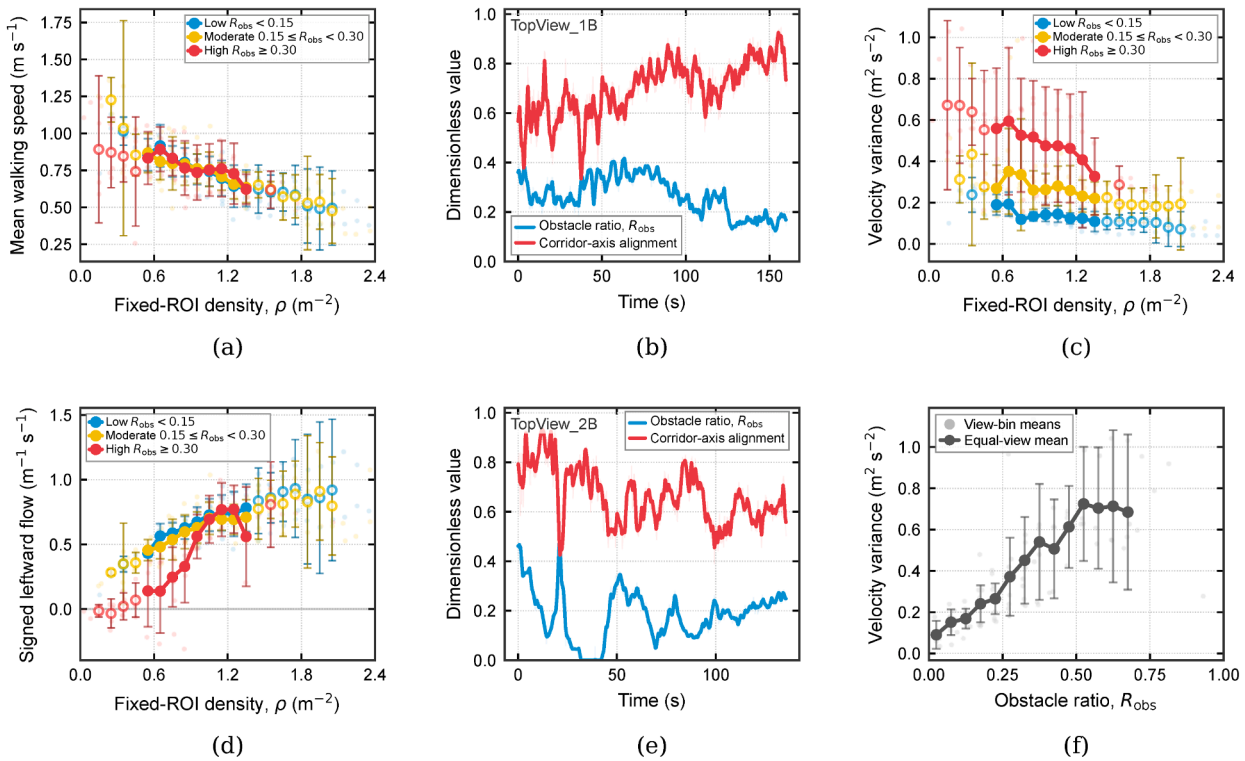


Fig. 5. Crowd-level relationships with the proportion of non-mainstream pedestrians. (a) Mean walking speed as a function of the fixed-ROI density for different obstacle ratios. (b) Obstacle ratio and corridor-axis alignment for TopView_1B. (c) Velocity variance as a function of the fixed-ROI density for different obstacle ratios. (d) Signed flow as a function of the fixed-ROI density for different obstacle ratios. (e) Obstacle ratio and corridor-axis alignment for TopView_2B. (f) Velocity variance as a function of the obstacle ratio. Points indicate mean values, and error bars denote 95% Student-t intervals across contributing camera views; in (f), gray points show view-bin means and black points show equal-view means.

3.4. Simulation responses of a continuum model

Finally, we use the continuum model to compare how Static, Counterflow, and Crossing pedestrians affect crowd-level motion under controlled conditions. Unlike the empirical analysis, where several pedestrian states may coexist, the simulations allow each non-mainstream state to be examined separately while keeping the remaining model settings controlled. The computational domain is $10\,\mathrm{m} \times 5\,\mathrm{m}$, as illustrated in Fig. 6(a), where Static pedestrians form a compact region near the boundary, Counterflow pedestrians form a longitudinal band, and Crossing pedestrians form a transverse band. The mean density is varied from 0.3 to $2.1\,\mathrm{m}^{-2}$, and the prescribed non-mainstream area fraction is varied from 0 to 0.40. The reference values of $K_{\min}$ are 0.62 for Static and 0.78 for Counterflow and Crossing. Because the prescribed area fraction ϕ does not necessarily equal the population fraction after density redistribution, we calculate the realized non-mainstream proportion as

$$R_{\mathrm{obs}}^{\mathrm{sim}} = \frac{\int \rho\phi \, \mathrm{d}A}{\int \rho \, \mathrm{d}A}. \tag{22}$$

Fig. 6(b)–(e) show that the three configurations produce different crowd-level responses. Mean walking speed decreases as the realized non-mainstream proportion increases in all three cases. At $\bar{\rho} = 1.2\,\mathrm{m}^{-2}$, the change per 0.10 increase in $R_{\mathrm{obs}}^{\mathrm{sim}}$ is $-0.061\,\mathrm{m\,s}^{-1}$ for Static, $-0.015\,\mathrm{m\,s}^{-1}$ for Counterflow, and $-0.034\,\mathrm{m\,s}^{-1}$ for Crossing, in Fig. 6(b). Signed flow also decreases for Static and Counterflow, by $-0.094\,\mathrm{m}^{-1}\,\mathrm{s}^{-1}$ and $-0.154\,\mathrm{m}^{-1}\,\mathrm{s}^{-1}$, in Fig. 6(c). Crossing shows the same downward tendency on the primary grid, but this response remains sensitive to grid resolution. The directional measures show clearer differences among the three states, as shown in Fig. 6(d), Static and Counterflow retain $S_{\parallel} = 1$, whereas Crossing reduces corridor-axis alignment because its prescribed motion is transverse to the corridor. Velocity variance increases most strongly for Counterflow, mainly through the longitudinal component, which reflects the coexistence of pedestrians moving in opposite directions in Fig. 6(e). The value of $K_{\min}$ mainly changes the magnitude of these responses in Fig. 6(f). At an area fraction of 0.30, increasing $K_{\min}$ from 0.40 to 1.00 increases mean walking speed by $0.129\,\mathrm{m\,s}^{-1}$ in Counterflow and $0.067\,\mathrm{m\,s}^{-1}$ in Crossing, while the change for Static is only $0.007\,\mathrm{m\,s}^{-1}$. The numerical checks indicate that the main conclusions are insensitive to the CFL setting: changing the CFL target from 0.30 to 0.45 changes the tested outputs by at most 1.8×10^{-6} in relative terms. Thirteen of the 15 grid comparisons meet the predefined 5% criterion, and the two exceptions are Crossing signed flow and Crossing velocity variance, which are therefore reported only as qualitative trends.

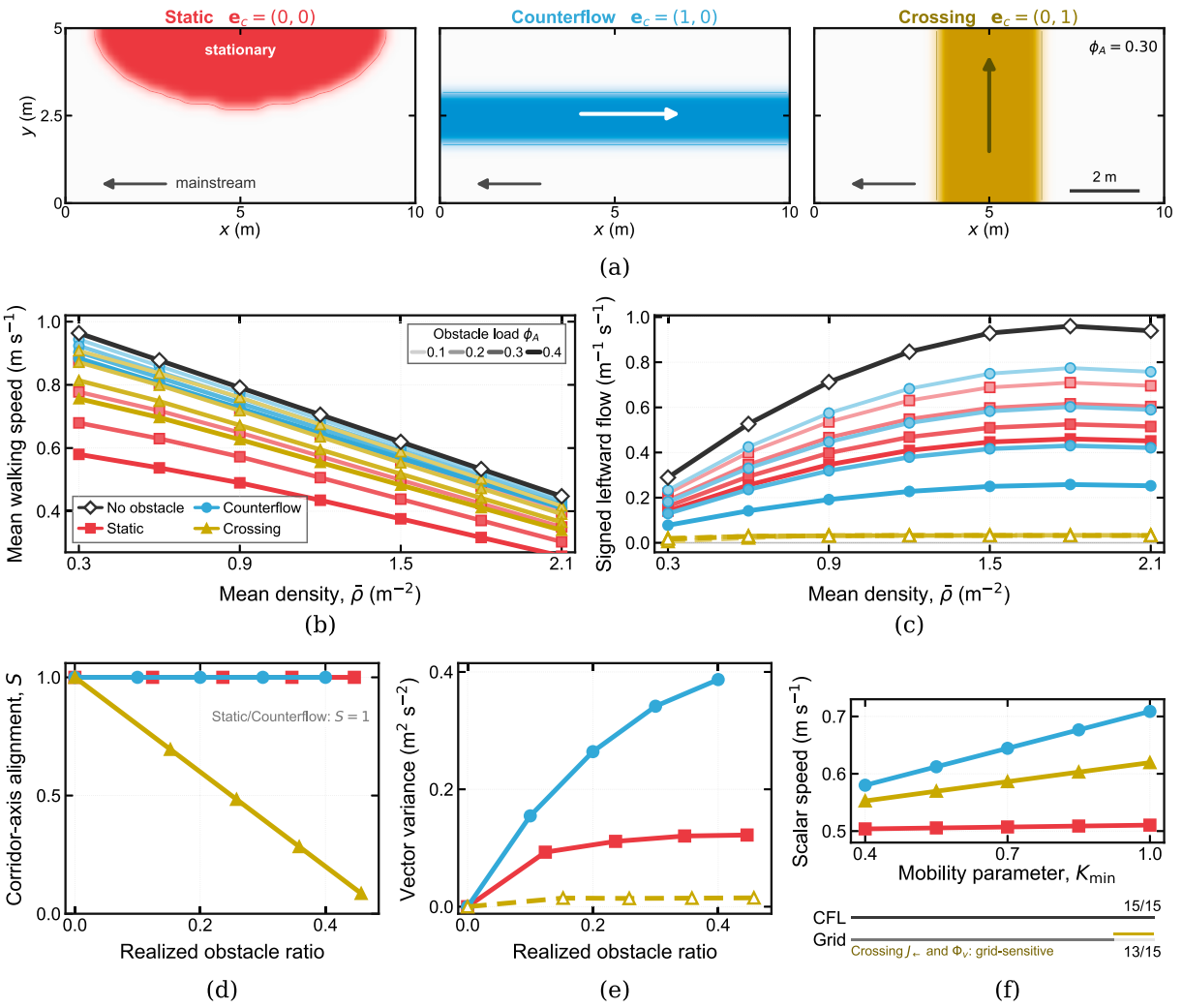


Fig. 6. Continuum-model responses for Static, Counterflow, and Crossing configurations. (a) Prescribed spatial distributions at a non-mainstream area fraction of 0.30. (b) Mean walking speed as a function of mean density across different area fractions. (c) Signed flow as a function of mean density across different area fractions. (d) Corridor-axis alignment as a function of realized obstacle ratio at mean density $1.2\,\mathrm{m}^{-2}$. (e) Velocity variance as a function of realized obstacle ratio at mean density $1.2\,\mathrm{m}^{-2}$. (f) Sensitivity of mean walking speed to mobility parameter at mean density $1.2\,\mathrm{m}^{-2}$ and area fraction 0.30; the bars below summarize the CFL and grid-resolution checks.

4. Discussion and conclusion

In this work, we combine empirical trajectory analyzes and continuum simulations to examine heterogeneous pedestrian motion at local and crowd scales. Four main findings emerge: (1) Static, Counterflow, and Crossing pedestrians exhibit distinct spatial and kinematic patterns, with consistent statistical signatures across camera groups, providing a common basis for characterizing non-mainstream motion. (2) Counterflow shows the clearest local differences, with the shortest time to closest approach among approaching pairs and the greatest local velocity variance. These patterns are consistent with opposing longitudinal motion, while squared lateral speed shows no clear differences among the three states. (3) After accounting for density, a higher non-mainstream proportion is associated with lower signed flow, lower corridor-axis alignment, and greater velocity variance, predominantly in the longitudinal component. Associations with alignment and velocity variance are most consistent, indicating that heterogeneous motion is better reflected in directional organization and velocity differences than in mean walking speed. (4) The continuum simulations distinguish the characteristic responses of the three states under prescribed motion assumptions: at the reference density, Static produces the largest reduction in mean walking speed, Counterflow produces the strongest increase in velocity variance and reduces signed flow, while Crossing reduces corridor-axis alignment.

This work also provides an empirical and methodological basis for diagnosing heterogeneous crowd motion and evaluating pedestrian models. First, the trajectory-based classification and local interaction measures establish a common framework for comparing stationary, opposing, and transverse motion, allowing encounter geometry, approach timing, and surrounding velocity heterogeneity to be examined separately. This distinction supports more precise interpretation of local interactions, particularly when a short approach time does not coincide with a small predicted separation (Olivier et al., 2012). Second, considering mean walking speed, signed flow, corridor-axis alignment, and velocity variance together distinguishes scalar mobility, net directional transport, and the organization of movement. These complementary measures can inform indicator selection for retrospective crowd assessment (Zhou et al., 2025) by revealing changes that are less apparent in conventional speed-density descriptions. Third, the empirical associations provide reference patterns for evaluating pedestrian simulations, while the continuum model enables controlled comparisons under explicit assumptions about motion composition. Together, they support model assessment beyond agreement with average speed or flow alone, including whether models reproduce differences in directional alignment and velocity heterogeneity (Duives et al., 2013). These contributions offer a basis for future model calibration and comparative evaluation of crowd-management scenarios.

Several limitations define the scope of these findings and suggest directions for future research. First, the pedestrian states are defined by trajectory-derived speed, direction, and predefined thresholds, without independent annotations of behavior or intention. Classification and several outcome measures also share velocity information, so their associations require a kinematic interpretation and do not independently establish causal mechanisms. Second, the empirical data come from a single public event, with two camera views covering the same physical scene. Although sensitivity analyzes support the stability of the principal alignment and velocity-variance relationships, additional settings are needed to establish their generality (Haghani and Sarvi, 2018). The combined non-mainstream proportion also does not separate the contributions of individual states or their interactions. Third, the centered classification windows use future observations, restricting the present implementation to retrospective analysis, while the 4 m pairing rule constrains lateral-separation profiles at large longitudinal separations. Finally, the continuum model prescribes motion directions and relative speeds and omits individual avoidance decisions and interactions among multiple non-mainstream states (Bode et al., 2019). Its Crossing estimates for signed flow and velocity variance remain sensitive to grid resolution and therefore warrant only qualitative interpretation. Future work could combine independent behavioral annotation, replication across transportation and event settings, classification using only current and past observations, and models that explicitly represent adaptive interactions. These extensions would help establish the framework's transferability and assess its potential for prospective crowd monitoring and operational planning.

CRediT authorship contribution statement

Wenfeng Yi: Writing – original draft, Validation, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation; **Wenhan Wu:** Writing – original draft, Visualization, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization.

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Data availability

The empirical trajectories are from the public dataset described by Dufour et al. (2025).

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.trc.2026.106094.

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